



Building a support framework for enterprise integration

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ARTICLE INFO

Article history:

Received 26 February 2012

Accepted 18 September 2012

Available online 9 October 2012

Keywords:

Enterprise architecture

Enterprise integration

Meta-methodology

ABSTRACT

Knowing 'what to do next' and what artefacts to use so as to achieve the desired future state of an enterprise is a recurring management conundrum. This paper describes a framework (and a possible application) aimed to support the efforts to achieve and maintain enterprises (or networks thereof) in a state where the flows of material and information are seamlessly integrated so as to not only achieve the internal and external synergy and agility required to survive but also to *thrive* in today's fast-evolving business environment. The framework is based on the integration of a structured repository containing architecture framework elements and a set of steps guiding the creation of specific enterprise engineering tasks. An outline of the theoretical foundation is followed by a high-level description of the requirements, architectural and detailed design descriptions of the proposed support framework and its potential application as a decision support system.

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1. Introduction

Typically, the enterprise engineering (EE) tasks involved in reaching and maintaining enterprise integration (EI) are on-going, complex, and often involve partially and/or ill-defined constraints. This is also because EI is indeed a multi-faceted concept involving consistency of data, dynamic allocation of resources and consistency in decision-making so as to address a coherent set of local and/or distributed objectives [1].

Thus, EE project managers are expected to constantly evaluate and understand the current situation, clarify and choose an optimal future state, grasp the scope and content of the change process required, gather knowledge of useful artefacts, and then select, model and communicate the 'way forward' to the organisation(s) involved.

Enterprise architecture (EA) holds the promise to help stakeholders manage the on-going EE effort in a consistent and coherent fashion, both within and between organisations, so as to achieve and maintain EI. However, EA itself is currently a rather new field and as such lacking agreed-upon consistent definitions. As a result, stakeholders and EE project managers are still rather confused as to (a) 'what needs to be done next' to achieve/maintain integration and (b) what artefacts (e.g. models, methods, frameworks, or combinations thereof) to use to model, evaluate, choose and communicate (a).

Current systems such as Executive Dashboards, based on Executive Information Systems principles as described by Volonino

and Watson [2], focus more on presenting existing data rather than on actively assisting the decisional effort. This paper describes interim results of the on-going research efforts aiming define a framework to support decision-making in EE projects by using the analysis of the interactions between project participants in the context of their lifecycles. The framework assists in the EE effort by eliciting the domain know-how of the project stakeholders and using it to discover and describe the relevant aspects of the EE project at hand – such as processes and their associated data, relations between participating entities, decisional aspects. Mainstream Architecture Framework (AF) elements and supporting tools are employed by the system in order to model the new knowledge thus created and help the participant organisation(s) understand and internalise it.

Note that due to scope and space limitations, the main aim of this paper is to explain the concepts and high-level structure of the proposed framework, rather than providing a full description. The interested reader can find such details in [3].

2. Requirements for the proposed framework

There are notable challenges in achieving EI (and implicitly, interoperability and compatibility as described in [4]). Panetto and Molina [5] summarize a substantial set of current issues in EI and interoperability featuring among others the need for design and engineering methodologies, reference model repositories (encapsulating reusable enterprise engineering (EE) models) and tools required to integrate business and manufacturing models. Currently, a considerable number of AFs offer artefacts aiming to satisfy the above-mentioned requirements; however, their terminology and coverage typically displays gaps and overlaps that

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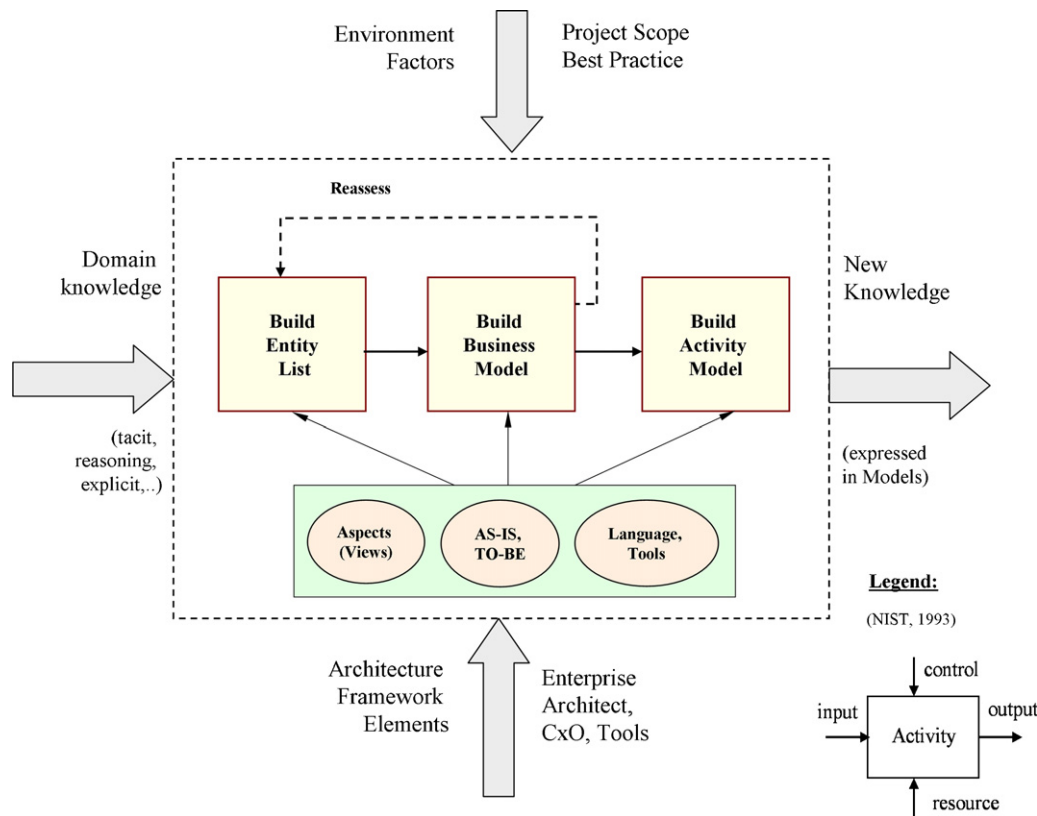


Fig. 1. Simplified meta-methodology concept (using an IDEF0-based [44] formalism).

make them confusing to intended users [6].¹ Most AFs also offer proprietary approaches and supporting tools – thus triggering the ‘vendor lock-in’ risk.

In addition, according to Molina et al. [7], new ‘killer’-type ICT-based applications and systems likely to become ubiquitous must support and satisfy several fundamental requirements, notably including EI and interoperability, human–software/hardware integration, cooperation and agility.

Past and current research of the author has tried to define the guidelines for an approach tackling the above-mentioned needs in an *integrated* manner by providing a framework integrating a repository of AF artefacts supporting EI/EE/EA and a method to create specific EE tasks based on a set steps and the selection of artefacts from the above-mentioned repository. It is hoped that this approach will help address the methodology, modelling and reusability aspects of the above-mentioned challenges identified in the EI domain.

3. Theoretical base and components of the support framework

The foundation of the proposed support system originates in research attempting to assess the feasibility and define the principles of a *method to create methods* (‘meta-methodology’) guiding specific projects in the Collaborative Network (CN) and Virtual Organisation (VO) area [8,9]. As the name implies, the main result of the meta-methodology application to a specific project is a method typically expressed in one of more activity models depicting tasks necessary to accomplish that project. The research has defined several steps required to achieve the above-mentioned outcome: thus, the project stakeholders must first understand the problems of the current state, then reason about possible ways to

solve them and subsequently select the optimal solution for the project at hand. This involves modelling various viewpoints using appropriate views, formalisms, reference models (RMs) possibly grouped in modelling frameworks (MFs). The meta-methodology assists stakeholders in using their (often tacit) domain knowledge (Fig. 1 left) to infer new facts by following a guided process of business modelling. The newly created knowledge is explicit (expressed in the models created) and used primarily to accomplish the project. However, it will typically be also internalised by other stakeholders and reflected in future decision-making [10], thus enriching and completing the corporate knowledge lifecycle. The meta-methodology has been tested in several case studies, e.g. [9,11].

3.1. A step-by-step method to construct methods

As can be seen from Fig. 1, currently the meta-methodology comprises three major stages and a set of sub-steps. In the first stage, the user is prompted to create a list containing entities of interest to the project in question, making sure to include project participants, target entities (organisations, other projects being built/managed by the project at hand) and importantly, the EE project itself. The second stage comprises the creation of business models showing the relations between the previously listed entities *in the context of their lifecycles*, i.e. illustrating how entities influence each other within each life cycle phase. The third stage assists the user in inferring the set of project activities by reading and interpreting the previously represented relations for each life cycle phase of the target entities. The activities must be detailed to a level deemed as comprehensible (and thus usable) by the intended audience. This task is assisted by other models and artefacts built and adopted during the second stage.

The first meta-methodology sub-step calls for the selection of suitable aspects, or viewpoints reflecting stakeholder concerns

¹ Previous research [7,8] has attempted to address the lack of a commonly agreed-upon terminology leading to reduced AF acceptance.

(ISO/IEC 2005; [12]) to be modelled in each stage (note that the life cycle viewpoint representation is mandatory and essential to the meta-methodology). The selection of a MF is also recommended, as MFs typically feature structured collections of views that can be used as *checklists* of candidate aspects and their intended coverage. This sub-step also requires the identification and resolution of any aspect dependencies. The second sub-step asks the user to determine if the present (AS-IS) state of the previously adopted views needs to be shown and whether the AS-IS and TO-BE (future) states should be represented in separate, or in combined models. The third sub-step requires the selection of suitable modelling formalisms and modelling tools for the chosen aspects.

More details on the meta-methodology stages and sub-steps are presented in Section 4.3. It is to be noted that all the above-described stages and sub-steps have underlying logic guiding the user in the decision-making process. A simplified sample subset of this logic (refined and implemented in the form of rules and facts in the practical application of the support framework) is presented in Section 4.2.

Importantly, research and testing have determined that although originating in the CN/VO area, the meta-methodology is in fact applicable to *any* EE project type, due to its open character and neutral approach towards all major AFs and enterprise modelling approaches [3].

3.2. A structured repository of AF elements

The meta-methodology concept relies on creating and interpreting models depicting various aspects of the EE project participants. This modelling effort requires the selection of methods, languages, tools, MFs and RMs according to specific project requirements and best practice. Since such artefacts are typically structured in AFs, the question is how to find suitable sets of AF components for specific projects. A possible answer is to construct *ranked lists* of suitable AF elements and ask the stakeholders to pick the most appropriate ones in view of their domain knowledge (which is a *required* input to the meta-methodology – see Fig. 1 left).

For this to be possible, AFs must first be decomposed in elements assessed on their scope, integration, dependencies, etc. and organised in a coherent collection using a set of consistent criteria. Thus, mainstream AFs have been (and still are) assessed and decomposed in respect to ISO15704 Annex A (GERAM,² see [13] for details). The reason behind the choice of reference AF is twofold: firstly, GERAM attempts to be neutral in respect to all assessed AFs and secondly, it contains most of the components and viewpoints necessary for AF element classification.

PERA³ [14], GRAI-GIM⁴ [15,16], CIM-OSA⁵ [17], ARIS⁶ [18], Zachman⁷ [19] and DoDAF⁸ [20] have been decomposed using GERAM [21,22] (see Fig. 2, top). Other AFs, such as TOGAF⁹ [23], TEAF¹⁰ [24] and FEAF¹¹ [25] are currently being added to the repository. The collection of AF elements obtained in this way (see Fig. 2, bottom) has been structured using criteria such as type, family. The result of this arrangement is referred to hereafter in this paper as the ‘structured repository’ (SR).

4. Design of the support framework

4.1. Knowledge and decision making in enterprise engineering and integration

Knowledge is not only present in all organisations in various forms (explicit, tacit, descriptive, procedural, reasoning) but also continuously produced (or *converted* [26]) as a consequence of knowledge-intensive decision-making processes [27]. Knowledge is an enabler of organisational *agility*, ensuring survival and competitiveness; thus, its capturing, representation, processing and the selection of supporting technology and infrastructure [28] is a major concern of management.

EE tasks present as part of the EE effort often display a semi-structured character due to insufficient information and the inherent complexity of organisational change processes. This may require EE project managers to make ‘semi-programmed’ decisions [29] that are difficult to encode in a program, but can be facilitated by a decision support system (DSS) [30].

This obvious need for decision-making guidance and support in EE has led to the idea of materialising the knowledge acquisition, structuring and transformation capabilities of the meta-methodology presented in Section 3.1 in the form of a support framework for EE (and a possible DSS practical application) that would help stakeholders such as enterprise architects, project managers and Chief Information Officers identify problems, find suitable solutions and define change processes to implement them.

EE decision-making involves problem solving in addition to using well-defined procedures. This mandates a cooperative and interactive support framework, allowing the users to address the part of the problem that cannot be structured and to use their own insights to modify and/or refine the solutions proposed by the system [31]. In this specific case, the users must be able to review and accept/override the ordering in the ranked lists of elements suitable for the EE problem. Interactivity and cooperation promote acceptance of the approach proposed by the framework; equally important, they put to use natural knowledge management skills and talents that cannot be programmed into a machine [27].

Finlay [32] has defined some basic requirements for DSS, which are reflected in the proposed framework’s functionality: help detect existing problems, model a problem situation to clarify it, provide the means to consider various options (e.g. via simulation) and help with implementation of change (by assisting in the modelling of the TO-BE states). The framework attempts to provide assistance in all essential decision-making phases [29]: intelligence gathering (via AS-IS modelling), design (via TO-BE modelling of several scenarios), choice (via simulation) and review (via the analysis of scenario results).

Other typical DSS requirements such as robustness and ease of use and control [33] can be satisfied here by using tested off-the shelf (but highly configurable) components providing most necessary services, such as web-based shells. This allows focusing the effort towards the knowledge repository development and user interface improvement, rather than on the intricacies of implementation. The knowledge captured by the system should be easily accessible for debugging and upgrading – thus preferably stored in plain text format.

The framework must also allow the user to bookmark important decision points so as to be able to backtrack if the decisions made have led to unsatisfactory outcomes. This would allow the analysis of various scenarios and cycling until an acceptable solution is obtained [34].

4.2. The architectural design of the proposed framework

The close connection between EE decision-making and knowledge management has suggested a rule-oriented knowledge-based

² The generalised enterprise reference architecture and methodology.

³ The Purdue enterprise architecture framework.

⁴ Graphs with results and activities interrelated.

⁵ Open system architecture for computer integrated manufacturing.

⁶ Architecture for information systems.

⁷ The Zachman architecture framework.

⁸ Department of defence architecture framework.

⁹ The open group enterprise architecture framework.

¹⁰ Treasury enterprise architecture framework.

¹¹ Federal enterprise architecture framework.

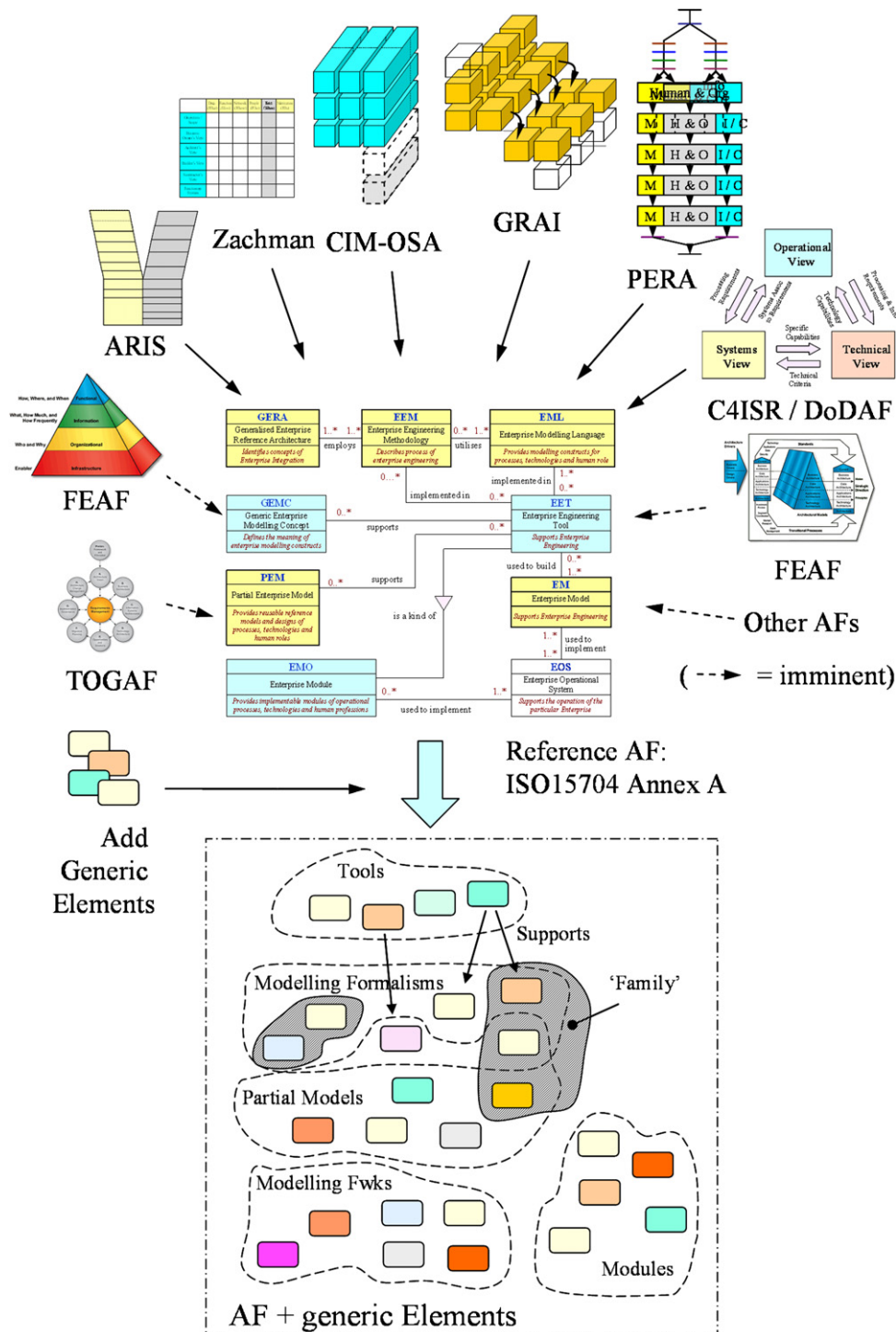


Fig. 2. Assessment of mainstream AFs using GERAM.

paradigm [27] for the proposed framework and its practical application as a DSS. The adopted structure of the system builds on previous mainstream research in this area. Thus, essential elements of knowledge-based DSS described by Sprague [35], Sprague and Carlson [36] and Marakas [37] are present in this design – such as a rule-based knowledge base where database elements are represented as facts and rules, a model base (populated by RMs of the AFs or abstracted from previous projects) and a dialog generation mechanism (possibly provided by an inference engine and its associated user interface). Hence, the DSS based on the proposed

framework will seek to apply best practice using rules/ranking facts and to consider user insights using run-time asserted facts. This will construct a set of solutions ordered by suitability and presented in a form that will assist the decision-making effort [38].

Fig. 3 presents the knowledge base approach adopted for the SR described in Section 3.2, containing rules for element selection (e.g. via pattern matching) and ordering (for ranked lists), fixed facts (e.g. AF element representations) and other rules containing the logic necessary for system operation. Typically, rules take the form of IF/THEN statements (their implementation depending on the

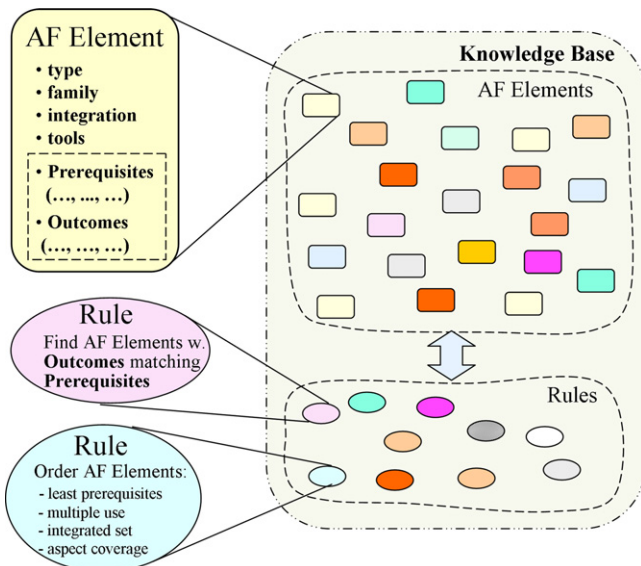


Fig. 3. Structured repository in knowledge base format.

specific inference engine used). For example, the rule to decide whether to model the present state (AS-IS) in an EE project could take the following form:

IF((TO-BE.obtained.from.AS-IS) OR (AS-IS.not_understood))
THEN(model.AS-IS) (1)

Thus, the 'model the present state' (model.AS-IS) rule will fire if the user asserts the fact that the AS-IS is not understood or the fact that the TO-BE will be obtained from the AS-IS. Another example:

IF(undecided.TO-BE) THEN (several.TO-BE) (2)

and, as a likely consequence:

IF(several.TO-BE) THEN(separate.AS-IS.TO-BE); (3)

Thus, if the stakeholders do not have a clear and common view of the future state of the enterprise and assert this at run-time (AS-IS_not_understood in (1)), then several TO-BE states will be

required (see (2)) and it is likely that they will need to be shown separately from the AS-IS state (see (3)). Note that although (3) is a somewhat typical consequence of (2), the user is allowed to override (3) if there is a compelling reason to do so. Generally, various levels of interactivity can be provided so as to match the flexibility needs and skill level of the user.

The stages and sub-steps of the underlying meta-methodology (see Fig. 1) have been incorporated as rules in the knowledge base, as shown in Fig. 4 bottom-left. Thus, in the current form, the meta-methodology stages, sub-steps and logic required for their proper operation are modelled by rules, the AF elements are represented by fixed facts while the user motivations/decisions (accept/override system recommendations) and other inputs are modelled via run-time asserted facts. As expected, the solutions are to be built by (a) rules implementing stage and sub-step logic, firing according to run-time asserted facts and (b) pattern-matching routines, ensuring that all dependencies of the fixed facts representing recommended and/or selected AF elements are satisfied.

4.3. Detailed design: main steps of the proposed support system

Due to space constraints, detailed design descriptions have been limited to the components reflecting the original meta-methodology stages and sub-steps described in Section 3.1 and Fig. 1. The level of detail has been kept low and plain text has been used to describe rule components so as to reduce complexity and help understanding. Once again, the interested reader is directed to [3,9] for details.

4.3.1. Stage One: create a list of entities relevant to the EE project

System explanation/recommendation (E/R): At this early stage, type and granularity consistency of the entities relevant to the project is not important. Thus, the set may contain documents, entire organisations, individuals, policies, etc. The *target* organisation(s) (i.e. that are being created or evolved) and the EE project itself *must* be included in the list.

4.3.2. Stage Two: business models of entity relations in a life cycle context

E/R: Represent the interactions between the entities listed, in the context of their life cycles (i.e. for each life cycle phase of the

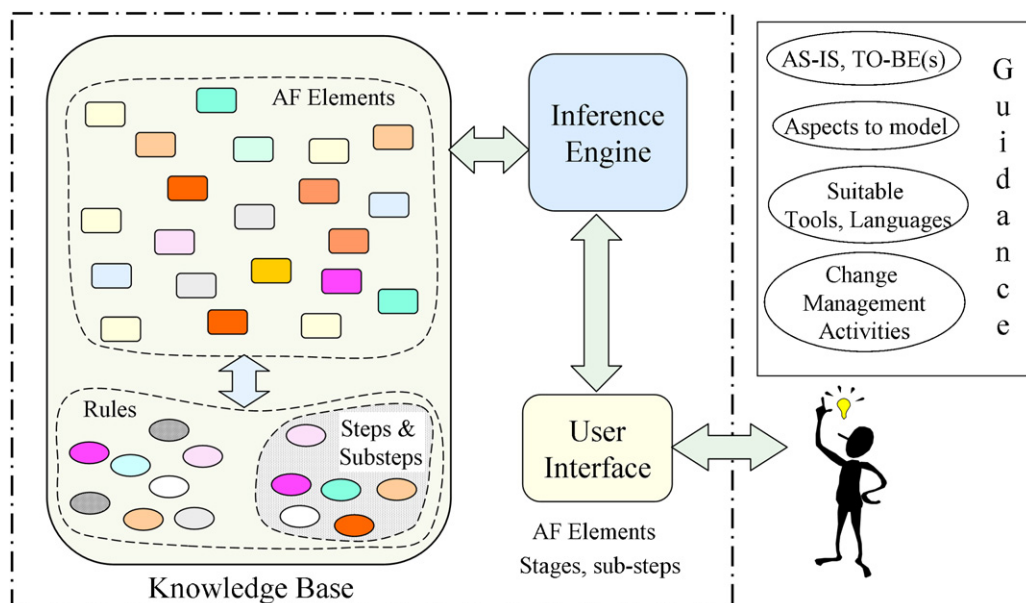


Fig. 4. Decision support framework architecture and possible outcomes.

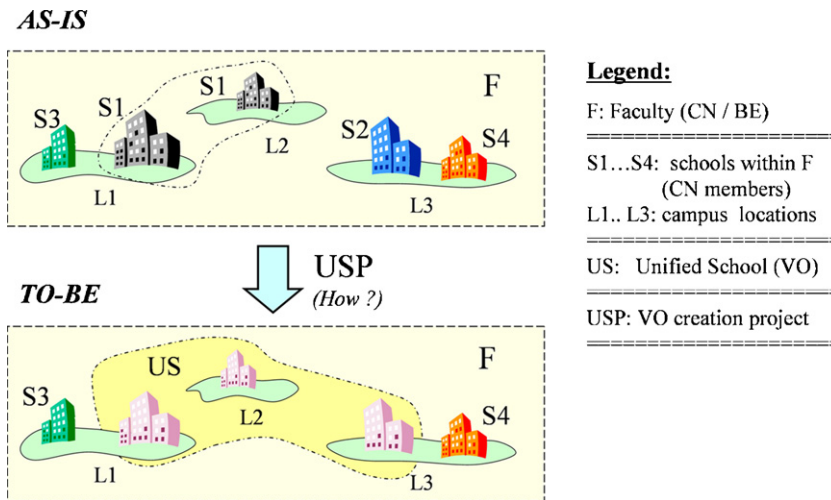


Fig. 5. AS-IS and TO-BE states of the enterprise engineering project.

target entities). Focus on the target organisation(s) and on the EE project. Restrict the representation of each entity to the extent of life cycle phases that is relevant to the project in question (to keep model complexity low). In view of the knowledge gained while building the models, re-assess the relevance of each entity to the EE project and the need for its presence in the models.

4.3.3. Stage Three: the set of activities describing the EE project accomplishment

E/R: Create a set of activities for each life cycle phase of the EE project and of the entities to be created/evolved by it. Describe the inputs, outputs, controls, resources, etc. needed for each activity using the interactions detailed in the business models created in Stage Two. Decompose the set of activities using views chosen in previous Stages, contained in the selected MF and in the knowledge base (in that order).

4.3.3.1. Sub-step A: (select relevant aspects and an MF). *E/R:* Unless explicitly mandated, this sub-step need not apply to Stage One. Re-use of aspects selected in previous Stages is recommended (system to build and display list, user to confirm/override).

Rules: Life cycle aspect mandatory. Functional aspect mandatory in Stage Three. Select an MF (a) as mandated by user, (b) according to existing skills, or (c) containing most of the aspects required by the project scope. Some typical aspects: management (or control)/service (or production, depending on enterprise's profile), human extent, decisional, organisational, functional, resources, informational, stakeholder, concern, hardware/software. Attempt to recommend an MF according to project scope; if none found suitable use the MF of GERA¹² (as currently having the most usable artefacts); recommend complementary views if the selected MF lacks them.

Stakeholder input (SI): Accept/override recommendations.

Rules: Check dependency and overlap of selected aspects, warn and confirm with user if overlaps detected and/or additional aspects required. Record decision, set bookmarks to enable backtracking to decision junctions.

4.3.3.2. Sub-step B: (AS-IS required? Type of AS-IS and TO-BE representation). *E/R:* Unified AS-IS/TO-BE representation is recommended in Stage One. In Stage Three, use the same representation type as in Stage Two unless otherwise mandated.

Rules: Apply this sub-step for all aspects selected in Sub-step A. TO-BE representation is mandatory. AS-IS must be represented if there is no clear common understanding of the present state by stakeholders, or if the TO-BE state must be evolved from the AS-IS. Represent AS-IS and TO-BE separately if there are obvious benefits (allow better critical analysis and more clarity for elaborate scenarios, etc.).

SI: Accept/override recommendations, make decision.

Rules: Record decision, set bookmarks.

4.3.3.3. Sub-step C: (modelling formalisms, tools). *E/R:* Choose formalisms (modelling languages) that best support the aspects selected in Sub-step A and choose tools supporting these formalisms (ranked lists provided by the system). In Stage One keep things simple (e.g. use plain text editor).

Rules: Create ranked lists of suitable formalisms and tools for aspects selected in Sub-step A, using best practice criteria – e.g. used in previous Stages (priority criterion), potential reuse, aspect coverage factor,¹³ whether part of an *integrated* set (ranked on whether formally – e.g. via underlying metamodels – or not), etc. Ranking uses the properties displayed by all AFs present in the SR (see Fig. 3).

SI: Accept/override recommendations (e.g. particular language(s)/tool(s) preferred).

Rules: Perform dependency check for chosen elements, record decision, set bookmarks.

4.4. Implementation

Currently, the main research work is directed towards extending the SR and developing and testing the support system rules in real case studies until reaching a maturity level suitable for full-scale implementation. Nevertheless, prototypes of the system have been implemented using various expert system shells, such as JESS [39,40].

5. Case study: operation of the support framework

The behaviour of the proposed system can be tested before reaching the full implementation phase since most of the essential rules and facts are known at this stage. In this particular case, a

¹² Generalised enterprise reference architecture, a component of GERAM.

¹³ i.e. showing how well an AF element can represent a particular aspect for a given life cycle phase.

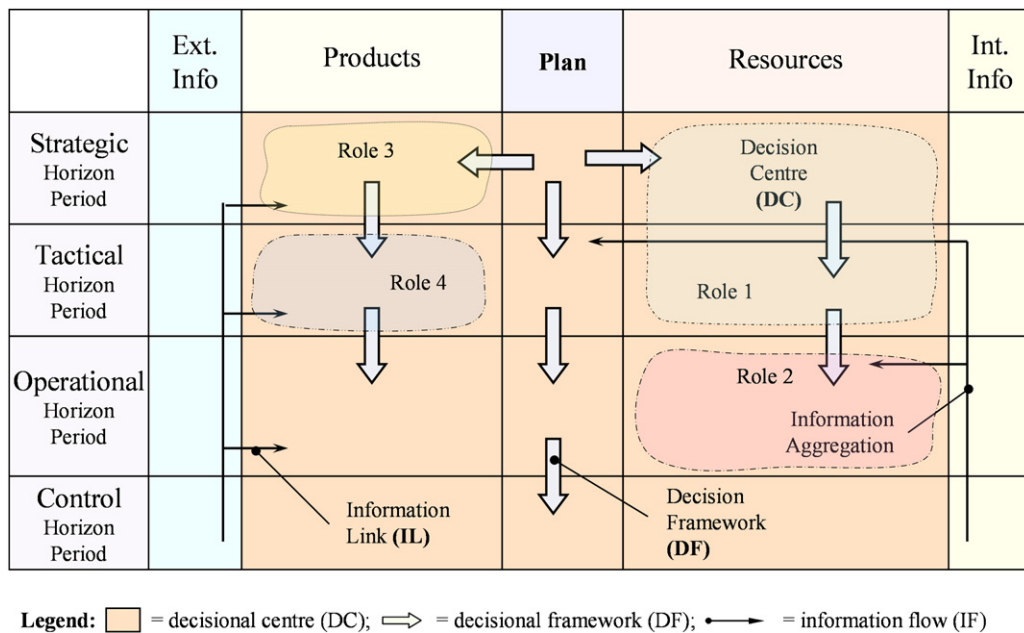


Fig. 6. GRAI-Grid modelling formalism for decisional aspects (mapped 'Roles' show a possible use of this formalism for organisational modelling).

facilitator familiar with the rules and facts present in the KB has performed the functions of the user interface and the inference engine, applying rules, asserting facts to reflect stakeholders input, building ranked lists and assisting in the creation, presentation, and interpretation of the results.

5.1. The specific enterprise engineering task

Faculty F of university U contains several schools, S1–S4. Schools S1 and S2 have the same specialisation but different organisational cultures and staff profiles. Geographically, school S1

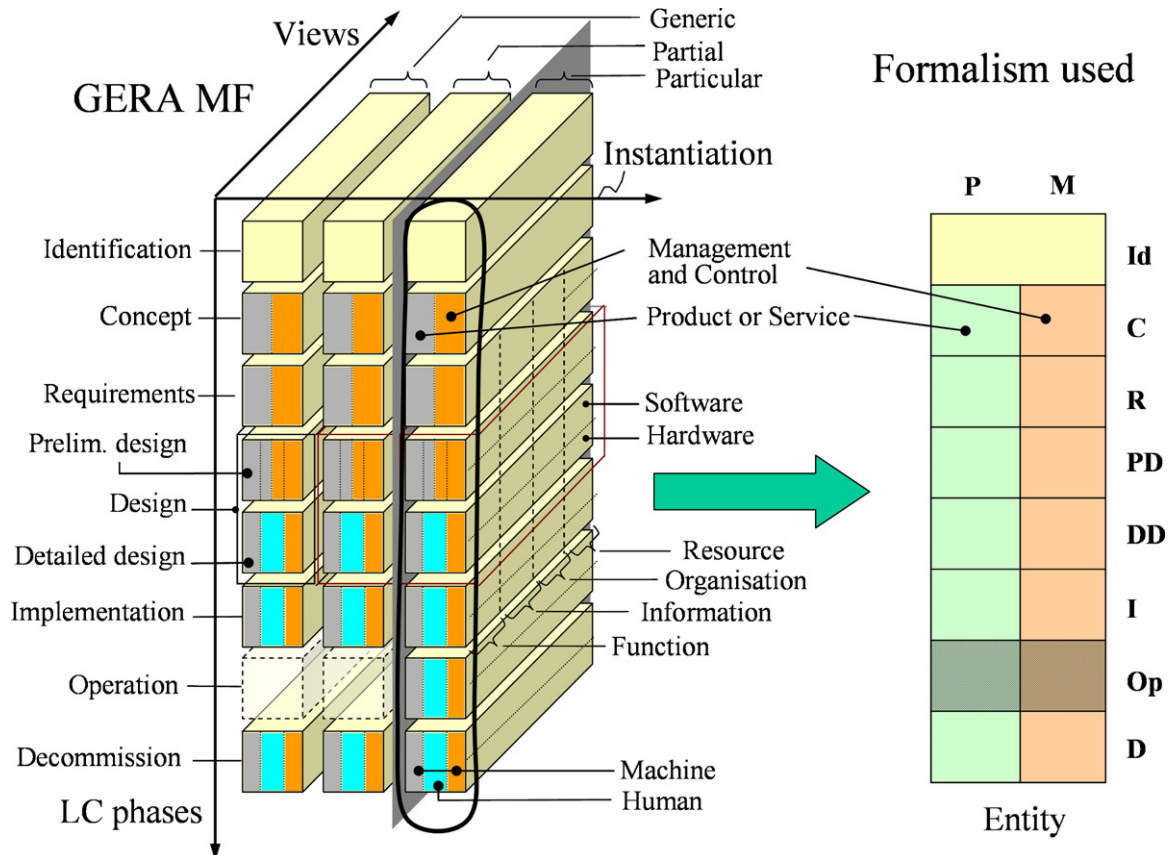


Fig. 7. Modelling formalism used for the management/service (or product) views of the business model. Note the strong life cycle context. Rectangles are placeholders for models.

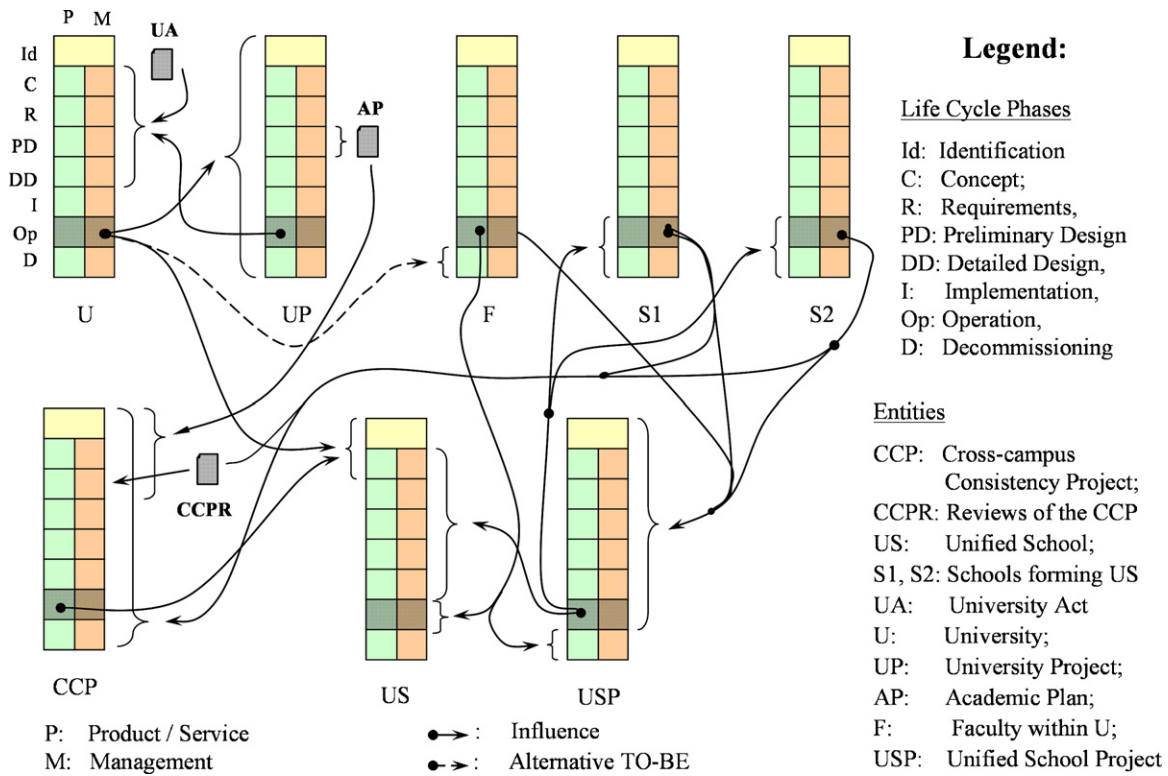


Fig. 8. Business model expressing entity roles in accomplishing the EE project.

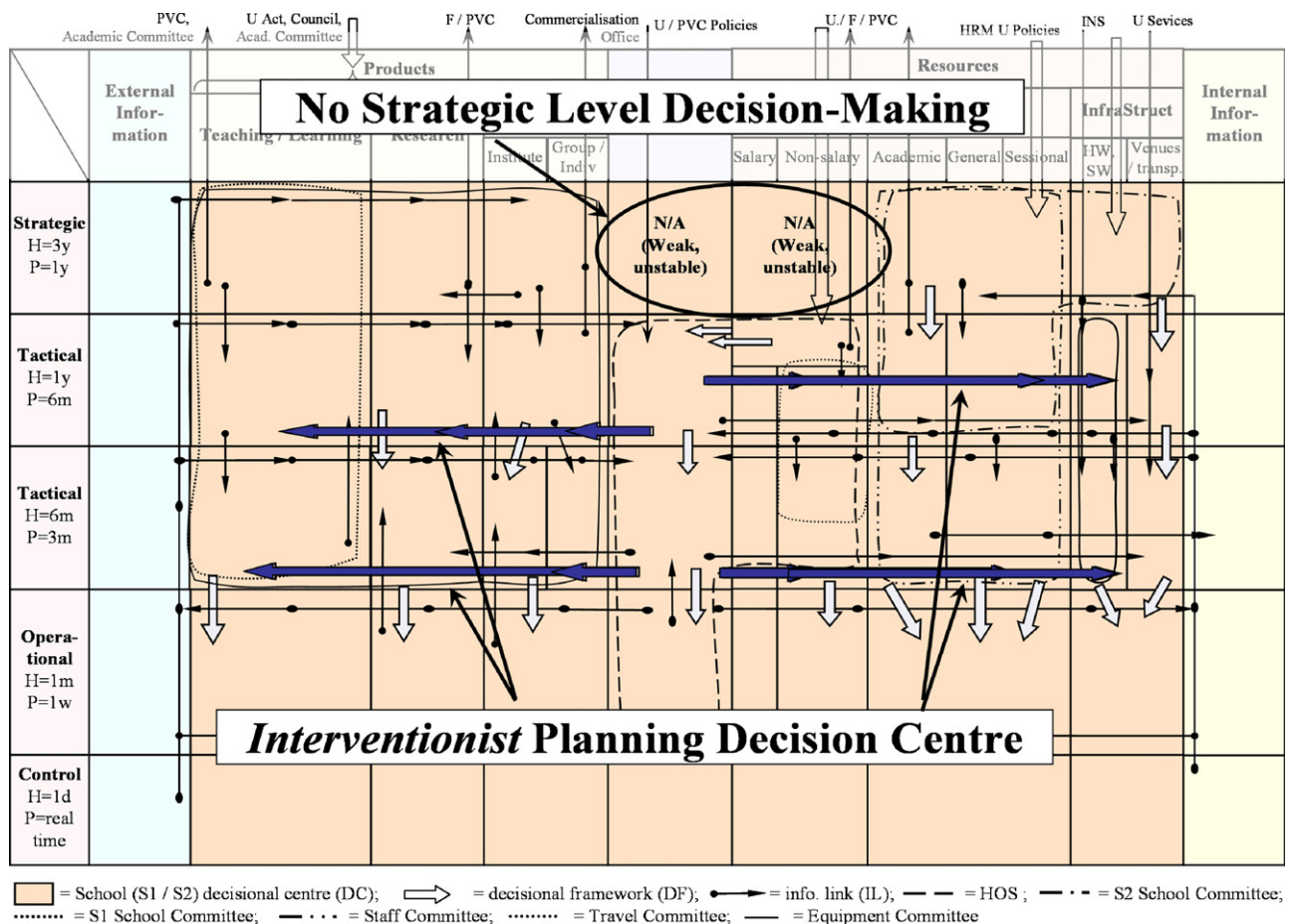


Fig. 9. Simplified AS-IS decisional/organisational model (DC text omitted for clarity).

is based at two campuses located at L1 and L2, while school S2 is based on a single campus located at L3 (see the AS-IS in Fig. 5). Although within the same faculty, and providing education in the same area, S1 and S2 are confronted with discrepancies in their products and resources – such as program content, allocated budget, scholarship conditions, academic staff profiles. This situation results in increased student administration and teaching maintenance costs, competition for resources, staff perceptions of inequity and it also projects a complex and confusing school image outside the faculty and university.

Previous experiences suggested that a cross-campus management layer steering a common approach to teaching and research would facilitate the coordination and consistency of the products delivered and of the resource allocation for the two schools. The proposed change process would make schools S1 and S2 appear as one indivisible entity to the outside environment, while preserving most of their current internal structure – thus, in effect forming a long-term VO (shown as ‘unified school’ (US) in the TO-BE area of Fig. 5).

Some typical hurdles in VO creation, such as trust building and agreement on the ICT infrastructure were to be alleviated by F and U in the role of *Breeding Environments* (BE) [41]. The main stakeholders of the project are the management of U, F, S1 and S2 (Heads of School, School Committees, Faculty Dean, Pro-Vice-Chancellor, etc.).

The stakeholders did not have a shared, agreed upon initial understanding of the present state of the schools, especially of the decisional and organisational aspects. Moreover, while the need for action and general direction had been identified and mandated by higher echelons, specific alternatives had not been specified. Thus, there was a need to understand the present situation,

identify the options available and devise a roadmap to achieve the selected solution.

5.2. Application of the decision support framework

5.2.1. Stage One: identify the entities relevant to the EE project

5.2.1.1. Sub-step A: (select relevant aspects and an MF if applicable). Stakeholder Decision (StD): No aspects selected (system recommendation accepted). **Stakeholder Motivation (StM):** Simplicity of the representation is paramount in obtaining a common understanding of the EE project scope.

5.2.1.2. Sub-step B: (AS-IS required? Separate or unified AS-IS and TO-BE?). (StD): Represent AS-IS. Use combined AS-IS/TO-BE representation.

(StM): No common understanding/agreed view on the present state of S1 and S2. No advantage in having separate but almost identical entity sets for AS-IS/TO-BE.

5.2.1.3. Sub-step C (modelling formalisms, tools). (StD): Choose plain text. Choose a text editor as ‘tool’ (accept system recommendation). **(StM):** Formalism and tool must preserve simplicity.

The entity list was built according to the selections made (entities legend in Fig. 8).

5.2.2. Stage Two: business models of entity relations in a life cycle context

5.2.2.1. Sub-step A (aspects, MF). (StD): No user preference for a particular MF; accept system default – GERA’s MF. Using the MF contents (see Fig. 7 left) as an aspect checklist, establish the

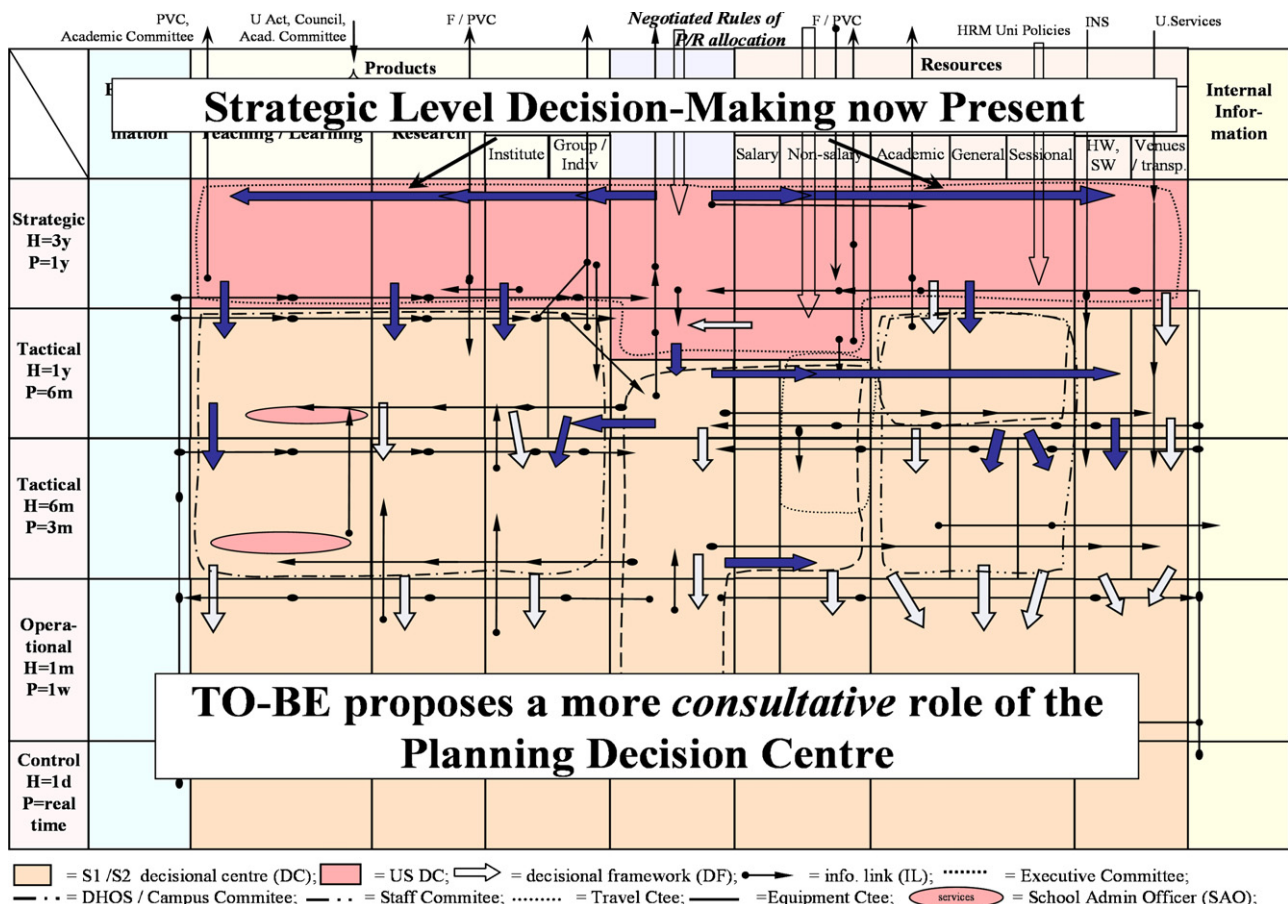


Fig. 10. Preferred TO-BE decisional/organisational model (DC text omitted for clarity).

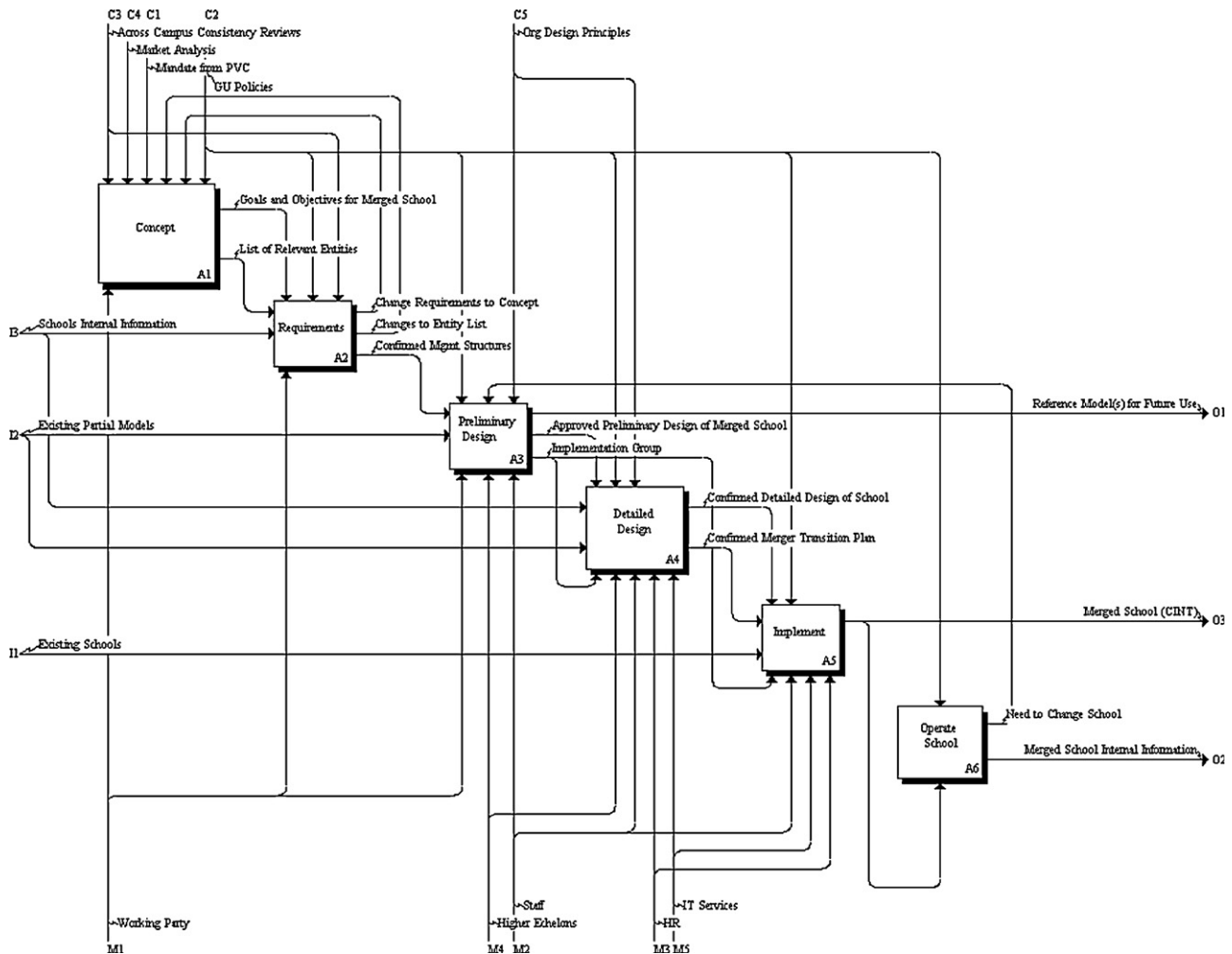


Fig. 11. Initial model of unified school creation and operation method: one activity per life cycle phase (Note: identification phase performed by higher echelons).

necessity for management/service, decision and organisational models. All aspects to be shown in the context of life cycle if applicable (see Stage Two definition).

(StM): Aspects selected as matching the scope of the project. Organisational aspect used to discern between the AS-IS and TO-BE states (see Sub-step B).

5.2.2.2. Sub-step B (AS-IS? Separate or combined AS-IS/TO-BE?). (StD): Represent the AS-IS state. For management/service and life cycle, represent AS-IS and TO-BE combined. Represent decisional and organisational views in *separate* models. (StM): No common understanding of the present S1 and S2 states. Separate AS-IS/TO-BE business models not necessary for the management/service aspect, but essential for organisational structures in order to distinguish between proposed TO-BE scenarios.

5.2.2.3. Sub-step C (modelling formalisms, tools). (E/R): Ranked lists of formalisms and tools with justifications are presented. For example in the decisional formalism area, GRAI-Grid [15] (Fig. 6) ranks highest for its specialisation and due to potential multiple use (decisional and organisational). Highest ranking tool: IMAGIM GraiSoft [42] for having *native* support for GRAI-Grid.

(StD): For management/service, accept system recommendation: a formalism based on GERA (Fig. 7 right). For decisional/organisational use GRAI-Grid. Tool: *simple graphical editor*.

(StM): No specific modelling language skills are available in the organisations participating in the EE project; hence, accept system

recommendations. However, tool recommendation *overridden* due to lack of specific skills and time constraints.

(E/R): Warning: due to override, models' syntax and consistency must be manually checked.

The models showing the selected aspects were constructed in view of the Stage Two system recommendations. The business model constructed for the management/service aspects (Fig. 8) shows several entities affecting various life cycle phases of the unified school US directly or via other entities. The main influence on US comes from the EE project (USP), which is decommissioned by the faculty F once US starts operating.

The AS-IS decisional model (see Fig. 9) has emphasized management problems such as interventionism of the Planning Decision Centre (DC) and a lack of effective strategy within the organisation. This is shown in Fig. 9 by Decisional Frameworks (DFs) flowing *horizontally* from Planning at several horizons. A lack of sufficient financial and decisional independence of S1 and S2 is also shown in the model by a lack of Planning DCs at strategic level and by DFs originating outside the schools.

The TO-BE decisional model attempted to address these problems by confining the authoritarian role of Planning to the strategic level and increasing the financial and decisional independence of US. This is reflected in a lesser number of horizontal DFs originating from Planning and in *negotiated* and *relocated* external DFs (see Fig. 10).

In this project, the TO-BE scenarios could only be distinguished by differences in organisational models. They were obtained by

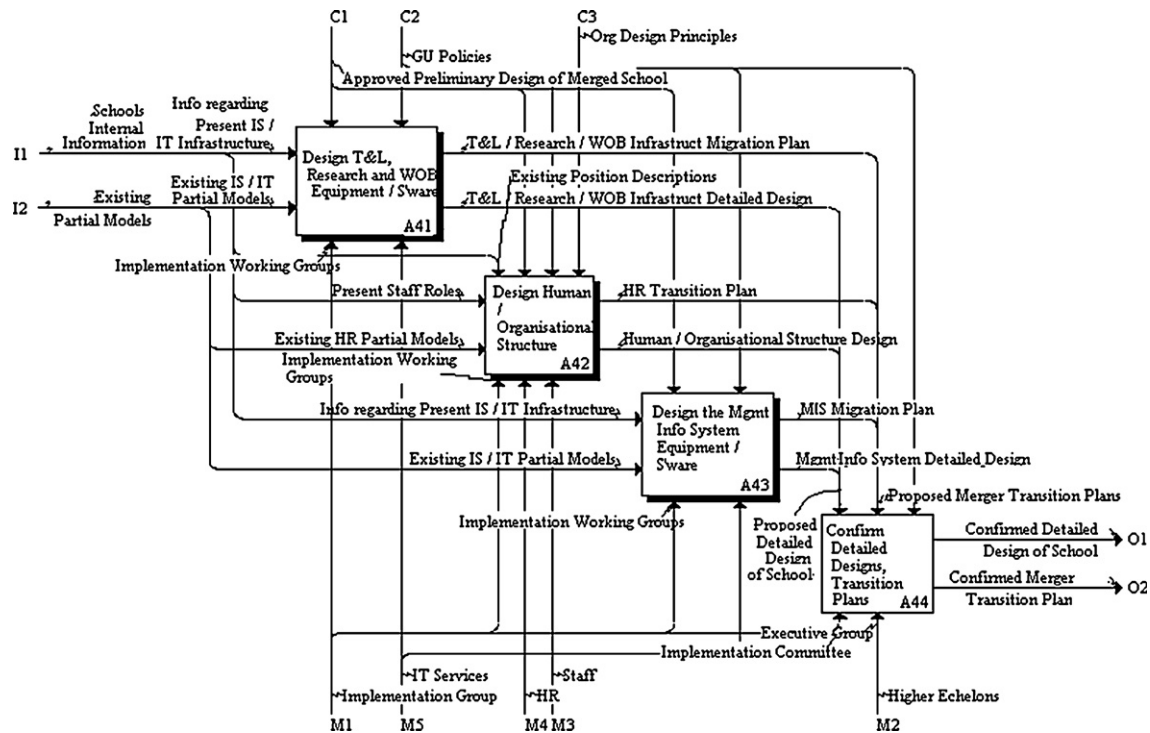


Fig. 12. A first decomposition of the detailed design activity (A4) using the management/product and human aspects.

allocating S1 and S2 human resources to DCs in the decisional model (see areas marked as various Committees and HOS in Figs. 9 and 10). Thus, having separate AS-IS and TO-BE decisional models here was essential.

5.2.3. Stage Three: the set of activities describing the EE project accomplishment

5.2.3.1. Sub-step A (aspects, MF). (E/R): Functional aspect is mandatory. Display previously used aspects, recommend use to detail the main functional model (and not to build separate models).

(Std): Select functional and life cycle aspects. Select management/service, decisional and organisational aspects. Add hardware/software from the views offered by the selected MF.

(StM): Functional and life cycle aspects are mandated in order to build the main deliverable (the activity model). Management/service, decisional and organisational aspects are directly addressed by the EE project and have also been previously used. Hardware/software aspect selected due to stakeholder domain knowledge (ICT infrastructure changes expected).

5.2.3.2. Sub-step B: (AS-IS required? Separate or unified AS-IS and TO-BE?). (E/R): Display and recommend Stage Two representation type (i.e. show AS-IS, and use combined AS-IS/TO-BE models).

(Std): Choose to represent AS-IS, and use a combined AS-IS/TO-BE representation.

(StM): System recommendation accepted. Stakeholders found no tangible gain in having separate views to justify the overhead of maintaining the consistency of several models.

5.2.3.3. Sub-step C (modelling formalisms, tools). (E/R): Present ranked lists of formalisms and tools with justifications. Example for functional top ranking formalisms: Unified Modelling Language (UML) Activity Diagrams (AD) belong to a set integrated by underlying metamodels [43] and have wide tool support. IDEF0 [44] is also suitable although no published

metamodel is available for its set (IDEF). Thus IDEF0 ranks lower than UML ADs. Proposed tools: Rational Rose [45] for UML AD, AIOWin [46] for IDEF0.

(Std): Select the IDEF0¹⁴ language and the AIOWin modelling tool.

(StM): System recommendation overridden based on domain knowledge: IDEF0 modelling skills and the AIOWin modelling tool are available within the participant organisations.

The activity model has been created next. Fig. 11 shows the initial phase, where one activity is created for each life cycle phase of the target entity. Note that often, the chosen modelling formalism helps reason about the aspects needing to be modelled. For example, if using IDEF0, the system can ask the user to give consideration to what is used in each activity (inputs), what controls it (controls), what is produced by the activity (outputs) and who and/or what executes it (resources). This information is obtained from the interactions shown in the business model (Fig. 8), other models (e.g. decisional) and stakeholder domain knowledge.

In this project, each activity has been decomposed down to a level where it was understood (and thus usable) by the envisaged audience, using aspects previously selected. For example, in Fig. 12, the management vs. product/service and human role aspects have been used for an initial decomposition of the detailed design activity shown in Fig. 11.

Not all aspects are always relevant or required. Early life cycle phases (e.g. identification, concept) require few or no aspects, while the human aspect may only be visible/relevant in the preliminary/detailed design phases, such as illustrated in Fig. 7, left.

Further detailed descriptions of the activity model creation and structure are beyond the space allocated and the scope of this paper; they are however available in [3,47].

¹⁴ A very brief summary of the main elements of IDEF0 is shown in Fig. 1, lower right.

6. Conclusions and further work

The EE effort aiming to achieve and maintain EI is a complex, on-going, and often ill-defined undertaking. The stakeholders need all the help they can get in order to understand the present state of affairs and turn the rather generic 'need for integration and continuous improvement' into concrete change steps towards achieving and upholding these aims. In addition, the multitude of AF artefacts on offer is confusing to the potential user and compounds rather than solve the problem.

The proposed support framework and its practical DSS application attempt to address these issues by offering guidance towards the creation of an enterprise modelling method specific to the EE project at hand. They also provide for model reuse and identification of gaps and overlaps between the various mainstream AFs currently available. In particular, the DSS application aims to provide a hands-on, interactive environment where the stakeholders can analyse various scenarios and their outcomes so as to make an informed choice of the optimal path to achieve the desired TO-BE state.

Current research will be continued in several directions: incorporate elements of other AFs into the SR, refine/streamline the logic of the meta-methodology steps contained in the knowledge base facts and rules, evolve and mature the structure of the proposed decision framework built on the SR and the meta-methodology and test the support framework and DSS in further case studies.

References

- [1] F. Vernadat, Enterprise modeling and integration (EMI): current status and research perspectives, *Annual Reviews in Control* 26 (1) (2002) 15–25.
- [2] L. Volonino, H.J. Watson, The strategic business objectives method for guiding executive information systems development, *Journal of Management Information Systems* 7 (3) (1990–1991) 27–39.
- [3] O. Noran, Discovering and modelling enterprise engineering project processes, in: P. Saha (Ed.), *Enterprise Systems Architecture in Practice*, IDEA Group, Hershey, USA, 2007, pp. 39–61.
- [4] H. Panetto, Towards a classification framework for interoperability of enterprise applications, *International Journal of CIM* 20 (8) (2007) 727–740.
- [5] H. Panetto, A. Molina, Enterprise integration and interoperability in manufacturing systems: trends and issues. Special issue on enterprise integration and interoperability in manufacturing systems, *Computers in Industry* 59 (7) (2008) 641–646.
- [6] D. Chen, G. Doumeingts, et al., Architectures for enterprise integration and interoperability: past, present and future. Special issue on Enterprise Integration and Interoperability in Manufacturing Systems, *Computers in Industry* 59 (7) (2008) 647–659.
- [7] A. Molina, H. Panetto, et al., Enterprise integration and networking: challenges and trends, *Studies in Informatics and Control* 16 (4) (2007) 353–368.
- [8] O. Noran, A Meta-methodology for Collaborative Networked Organisations: Creating Directly Applicable Methods for Enterprise Engineering projects, VDM Verlag Dr. Müller, Saarbrücken, 2008.
- [9] O. Noran, A decision support framework for collaborative networks, *International Journal of Production Research* 47 (17) (2009) 4813–4832.
- [10] B. Kalpic, P. Bernus, Business process modeling through the knowledge management perspective, *Journal of Knowledge Management* 10 (3) (2006) 40–56.
- [11] O. Noran, A meta-methodology for collaborative networked organisations: a case study and reflections, in: P. Bernus, M. Fox, J.B.M. Goossenaerts (Eds.), *Knowledge Sharing in the Integrated Enterprise: Interoperability Strategies for the Enterprise Architect*, Kluwer Academic Publishers, Toronto/Canada, 2004, pp. 117–130.
- [12] ISO/IEC, ISO/IEC 42010:2007: Recommended Practice for Architecture Description of Software-Intensive Systems, 2007.
- [13] ISO/IEC, Annex A: GERAM, in: ISO/IS 15704:2000/Amd1:2005: Industrial Automation Systems – Requirements for Enterprise-reference Architectures and Methodologies, 2005.
- [14] T.J. Williams, The Purdue enterprise reference architecture, *Computers in Industry* 24 (2–3) (1994) 141–158.
- [15] G. Doumeingts, La Methode GRAI, University of Bordeaux, Bordeaux, France, 1984.
- [16] G. Doumeingts, GIM, Grai Integrated Methodology, in: A. Molina, A. Kusiaka, J. Sanchez (Eds.), *Handbook of Life Cycle Engineering – Concepts, Models and Methodologies*, Kluwer Academic Publishers, Dordrecht, 1998, pp. 227–288.
- [17] CIMOSA Association, CIMOSA – Open System Architecture for CIM. Technical Baseline, Ver 3.2., Private Publication, 1996.
- [18] A.-W. Scheer, *Architecture for Integrated Information Systems*, Springer-Verlag, Berlin, 1992.
- [19] J.A. Zachman, A framework for information systems architecture, *IBM Systems Journal* 26 (3) (1987) 276–292.
- [20] DoD Architecture Framework Working Group, DoD Architecture Framework Version 1.0, 2004 http://www.dod.mil/cio-nii/docs/DoDAF_v1_Volume_I.pdf, http://www.dod.mil/cio-nii/docs/DoDAF_v1_Volume_II.pdf (retrieved February 2007).
- [21] O. Noran, A mapping of individual architecture frameworks (GRAI, PERA, C4ISR, CIMOSA, Zachman, ARIS) onto GERAM, in: P. Bernus, L. Nemes, G. Schmidt (Eds.), *Handbook of Enterprise Architecture*, Springer Verlag, Heidelberg, 2003, pp. 65–210.
- [22] O. Noran, An analytical mapping of the C4ISR architecture framework onto ISO15704 Annex A (GERAM), *Computers in Industry* 56 (5) (2005) 407–427.
- [23] The Open Group, The Open Group Architecture Framework (TOGAF 8.1.1 'The Book') v8.1.1, 2006.
- [24] Treasury CIO Council, Treasury Enterprise Architecture Framework v.1, 2000 <http://www.eaframeworks.com/TEAF/teaf.doc> (retrieved February 2007).
- [25] The CIO Council, Federal Enterprise Architecture Framework (v1.1), 1999 https://secure.cio.noaa.gov/hpcc/docita/files/federal_enterprise_arch_framework.pdf (retrieved February 2007).
- [26] I. Nonaka, H. Takeuchi, *The Knowledge-Creating Company. How Japanese Companies Create the Dynamics of Innovation*, Oxford University Press, Oxford, 1995.
- [27] C.W. Holsapple, A.B. Winston, *Decision Support Systems: A Knowledge-Based Approach*, West Publishing, Minneapolis/St. Paul, 1996.
- [28] T. Davenport, L. Prusak, *Working Knowledge*, Harvard Business School Press, Boston, 1998.
- [29] H. Simon, *The New Science of Management Decision*, Prentice-Hall, Englewood Cliffs, NJ, 1977.
- [30] P.G.W. Keen, M.S. Scott Morton, *Decision Support Systems: An Organizational Perspective*, Addison-Wesley, Inc., Reading, MA, 1978.
- [31] E. Turban, *Decision Support and Expert Systems: Management Support Systems*, Prentice Hall, Englewood Cliffs, NJ, 1995.
- [32] P.N. Finlay, *Introducing Decision Support Systems*, Blackwell Publishing, Manchester, 1994.
- [33] J.D.C. Little, Models and managers: the concept of a decision calculus, *Management Science* 16 (8) (1970) B466–B485.
- [34] P. Hättenschwiler, Neues anwenderfreundliches Konzept der Entscheidungsunterstützung. Gutes Entscheiden in Wirtschaft, Politik und Gesellschaft, vdf Hochschulverlag AG, Zurich, 1999.
- [35] R.H. Sprague, A framework for the development of decision support systems, *Management Information Systems Quarterly* 4 (4) (1980) 1–26.
- [36] R.H. Sprague, E.D. Carlson, *Building Effective Decision Support Systems*, Prentice-Hall, Inc., Englewood Cliffs, NJ, 1982.
- [37] G.M. Marakas, *Decision Support Systems in the Twenty-first Century*, Prentice Hall, Upper Saddle River, NJ, 1999.
- [38] E.G. Mallach, *Understanding Decision Support and Expert Systems*, Richard D. Irwin, Inc., Illinois, 1994.
- [39] E. Friedman-Hill, *Jess in Action: Java Rule-based Systems*, Manning Publications Co., Cherokee Station, New York, NY, 2003.
- [40] E. Friedman-Hill, JESS – The Rule Engine for the Java Platform, 2007 <http://herzberg.ca.sandia.gov/jess/> (retrieved February 2007).
- [41] L. Camarinha-Matos, Foreword, in: L. Camarinha-Matos, H. Afsarmanesh, A. Ortiz (Eds.), *Collaborative Networks and their Breeding Environments* (Proceedings of the 6th IFIP Working Conference on Virtual Enterprises – PROVE 05), Springer Verlag, Toulouse/France, 2005, pp. 339–346.
- [42] GRAISoft, IMAGIM Software Product, 2002 <http://www.graisoft.com> (retrieved August 2002).
- [43] J. Rumbaugh, I. Jacobson, et al., *The Unified Modelling Language Reference Manual*, Addison-Wesley, Reading, MA, 1999.
- [44] NIST, *Integration Definition for Function Modeling (IDEF0)*, Computer Systems Laboratory, National Institute of Standards and Technology, 1993.
- [45] IBM, Rational Rose Family, 2007 <http://www.kbsi.com/Software/KBSI/AIOWIN.htm> (retrieved August 2007).
- [46] KBSI, AIOWin Software Product, 2007 <http://www.kbsi.com/Software/KBSI/AIOWIN.htm> (retrieved August 2002).
- [47] O. Noran, Using reference models in enterprise architecture: an example, in: P. Fettke, P. Loos (Eds.), *Reference Modeling for Business Systems Analysis*, Idea Group, Hershey, USA, 2006, pp. 141–165.



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